

SC Honors Geometry - Nappier
Homework 10.5 – Angle Relationships in Circles

Name: _____

Vertex ON the circle (tangent-chord).

1. A tangent-chord angle measures 68° . Find its intercepted arc.
2. A tangent-chord angle intercepts a 210° arc. Find the angle.

Vertex INSIDE (half the sum).

3. Two chords cross; the vertical arcs are 81° and 123° . Find the angle.
4. Two chords cross at a 95° angle; one vertical arc is 112° . Find the other arc.

Vertex OUTSIDE (half the difference).

5. Two secants: far arc 140° , near arc 48° . Find the exterior angle.
6. Two tangents: the arcs are 250° and 110° . Find the angle between the tangents.
7. An exterior angle of 28° has a near arc of 46° . Find the far arc.
8. A tangent and a secant: far arc 122° , near arc 50° . Find the angle.

Circumscribed angles.

9. A circumscribed angle's central angle measures 128° . Find the circumscribed angle.
10. Two chords cross at an angle of x° ; the arcs they intercept are equal. What kind of chords must they be, and what is x if each arc is 90° ?